

Qatar-4b

R-2544

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_211006 · created 2026-08-26T00:22:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459493.6925 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.054
Transit depth (Rp/Rs)^2	3.13 +/- 1.92 [%]
Orbital Inclination (inc)	84.62 +/- 2.15
Airmass coefficient 1 (a1)	1.66 +/- 1.68
Airmass coefficient 2 (a2)	0.17 +/- 0.57
Scatter in the residuals of the lightcurve fit is	123.93 %
Best Comparison Star	#2 - [433, 235]
Optimal Aperture	0.87
Optimal Annulus	6.75
Transit Duration (day)	0.074 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.177 $\pm$ 0.054 vs 0.138 (deviation 0.41 σ)
Duration fit vs published	0.074 d vs 0.0892 d (deviation 0.41 σ)
Mid-time O-C	4.5 min
Depth SNR	1.8
Residual scatter	123.93 %

Light curve

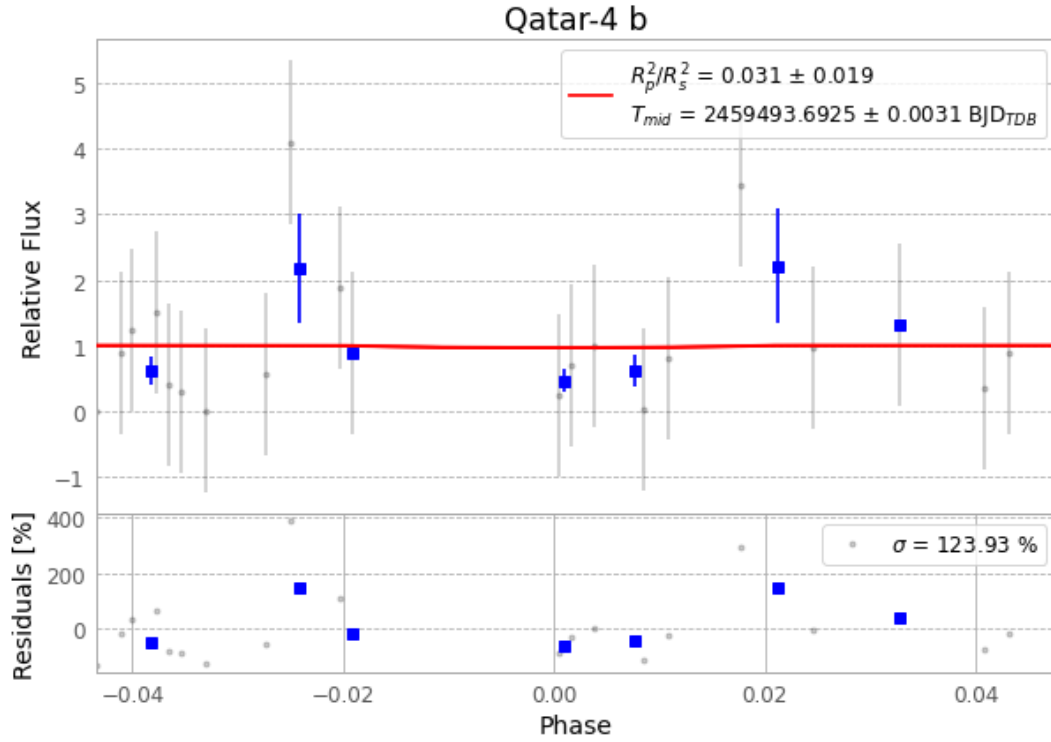


Figure 1 — FinalLightCurve_Qatar-4 b_2021-10-05.png (SHA-256 44c4e131b5ff4376...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [221, 416], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d4b5548d176fd7e7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

83 FITS frames (55 MB), Qatar-4211006023012.FITS → Qatar-4211006063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-211006.log (9c6bdd5335e2...), FinalLightCurve_Qatar-4 b_2021-10-05.png (44c4e131b5ff...), FinalLightCurve_Qatar-4 b_2021-10-05.csv (a40a1a5e9551...)

Compute resources

Wall time 195.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:22Z.